requirements are unnecessary and the resulting emissions data potentially confusing for investors.[932]

Further, commenters opposed the GHG emissions disclosure requirement because of the expected high compliance costs, which they believed the Commission had underestimated.[933] One commenter further indicated that, although the Commission had stated that many companies were already disclosing their GHG emissions, according to a number of studies, most registrants have not yet measured and reported their Scopes 1 and 2 emissions, let alone their Scope 3 emissions.[934]

Commenters also expressed concerns, in connection with registrants' disclosure of Scope 3 emissions, regarding compliance costs involving private companies, which comprise a large percentage of many registrants' value chains or joint ventures, and which, through the influence of those registrants, would be compelled to measure and report their GHG emissions for the first time.[935] Some of these commenters asserted that registrants would likely incur costs to renegotiate contracts with these third parties to obtain the GHG emissions data required to comply with the proposed rules.[936] Another commenter stated that third parties that are

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932 *See, e.g.*, letters from API; Chamber; and ConocoPhillips. According to commenters, confusion could result from the fact that the EPA's Greenhouse Gas Reporting Program ("GHGRP") requires the disclosure of emissions by individual source whereas the Commission's proposed rules would require the disclosure by company; *see also* discussion *infra* notes 2593-2595 and accompanying text. As noted in section IV.A.3, we estimate that approximately 365 registrants had an ownership stake in facilities that reported to the GHGRP in 2022; *see infra* note 2596and accompanying text.

933 *See infra* sections IV.C.3.b.ii and iii for more information on specific cost estimates provided by commenters.

934 *See infra* section IV.A.5c (citing statistics in the 2021 TCFD Status Report and a Moody's Analytics analysis of TCFD reporting of 2020/21 public disclosures showing that only 21% of North American companies and 19% of U.S. companies reported their Scopes 1 and 2 emissions and, if appropriate, their Scope 3 emissions).

935 *See, e.g.*, letters from API; Atlas Sand; Bipartisan Policy; Brigham Exploration; Chamber; ConocoPhillips; Independent Petroleum Association of New Mexico; and Iowa Commissioner of Insurance.

936 *See, e.g.*, letter from ConocoPhillips.

unwilling or unable to provide their GHG emissions to registrants could eventually be excluded from consideration for contracts to provide goods or services to registrants, which could diminish opportunities for these third-parties, which may often be smaller businesses.[937]

In addition, commenters stated that, even if registrants are already voluntarily disclosing their Scopes 1 and 2 emissions pursuant to the GHG Protocol, those registrants will incur an increased compliance burden if the Commission was to adopt the proposed GHG emissions disclosure requirement, because of differences between the Commission's proposed requirement and the GHG Protocol and the TCFD.[938] These commenters also shared many of the concerns about the proposed Scope 3 emissions disclosure provision discussed above, including the difficulties of collecting emissions data from third parties in its value chain, the unreliability of reported data stemming from third parties' lack of sophisticated data collection technologies and the use of proxy data to fill data gaps, and the absence of a fully developed and uniformly accepted methodology to report Scope 3 emissions. According to commenters, these concerns would increase compliance costs and raise a registrant's liability exposure so that the total cost of the Scope 3 emissions disclosure would likely exceed its benefit.[939] Because of the difficulties

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937 *See* letter from Soc. Corp. Gov.

938 *See id*. Specifically, the commenter noted that the proposed rules would require a registrant's organizational boundaries to be consistent with the scope of entities included in its consolidated financial statements, whereas the GHG Protocol permits a company to choose between an equity share, operational control, or financial control method. The commenter also noted that the Commission's proposed rules would require a company to disclose its GHG emissions both on a disaggregated and aggregated basis whereas the TCFD requires a company to disclose its Scopes 1 and 2 emissions, without specifying whether the disclosure must be on a disaggregated basis. According to the commenter, these differences could result in an increased compliance burden for a registrant. We discuss additional commenter input on these differences below.

939 *See id*; *see also* Bipartisan Policy; Brigham Exploration; Chamber; D. Burton, Heritage Fdn.; and the National Association of Convenience Stores (June 8, 2022).

and uncertainties involved in Scope 3 emissions disclosure, some commenters recommended that the reporting of Scope 3 emissions should remain voluntary.[940]

One commenter presented an alternative to the proposed GHG emissions requirement.[941] This commenter stated that, rather than adopting the proposed GHG emissions disclosure requirement, the Commission should "mandate reporting, on a standardized form, of emissions data that registrants are required to disclose publicly pursuant to other federal, state, or foreign regulations." This commenter also stated that the alternative set of rules "would, in effect, integrate the existing EPA reporting regime with the SEC's disclosure system in a manner that would be easier for investors and registrants to access and analyze."[942] This commenter further stated that approximately 40 foreign countries already require various forms of emissions disclosures, and that California and other states are considering the adoption of their own mandatory emissions reporting regimes.[943] According to this commenter, the alternative set of rules "would efficiently integrate, aggregate, and collate those disclosures on a single form available to all investors through documents provided to the Commission."[944]

Some commenters supported the proposed exemption from Scope 3 emissions reporting for SRCs.[945] Some commenters also supported exempting SRCs from the requirement to disclose Scopes 1 and 2 emissions because, in their experience, SRCs have not historically

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940 *See, e.g.*, letter from Airlines for America.

941 *See* letter from Joseph A. Grundfest, William A. Franke Professor of Law and Business, Stanford Law School (June 15, 2022) ("Grundfest"); *see also* letters from Joseph A. Grundfest, Professor of Law and Business (emeritus), Stanford Law School (Oct. 9, 2023); and Devon S. Wilson (Sept. 7, 2023).

942 Letter from Grundfest.

943 *See id*. As previously noted, California has since enacted a mandatory emissions reporting regime. *See supra* section II.A.

944 *See* letter from Grundfest.

945 *See, e.g.*, letters from D. Burton, Heritage Fdn.; J. Herron; ICI; Morningstar; and TotalEnergies.

tracked their GHG emissions and exempting SRCs from a GHG emissions reporting requirement would be consistent with a scaled disclosure regime for such issuers.[946]

Other commenters, however, opposed exempting all SRCs from the proposed Scope 3 emissions disclosure requirement.[947] Commenters stated that investors need climate-related disclosures from SRCs because SRCs are as exposed to climate-related risks as larger issuers, including risks stemming from their value chains.[948] Commenters also stated that because many large companies obtain climate-related data (e.g., Scopes 1 and 2 emissions data) from small companies in their value chains, exempting SRCs from climate-related disclosures could hamper larger registrants from accurately assessing their Scope 3 emissions.[949] Instead of, or in addition to, an exemption from Scope 3 reporting, some commenters recommended providing a longer transition period for SRCs.[950]

Some commenters recommended that the Commission exempt EGCs from the proposed rules, including GHG emissions reporting requirements, or at least provide them with the same accommodations as SRCs.[951] Commenters stated that the large compliance costs of the proposed

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946 *See, e.g.*, letters from BDO USA, LLP (June 17, 2022) ("BDO USA"); D. Burton, Heritage Fdn.; and Volta Inc. (June 15, 2022) ("Volta").

947 *See, e.g.*, letters from AGs of Cal. *et al*. (recommending requiring SRCs that have adopted transition plans with Scope 3 emissions reductions to report on those emissions); Amer. for Fin. Reform, Sunrise Project *et al*.; CalSTRS; CEMEX; Center Amer. Progress (stating that at a minimum, the final rule should establish a date in the future, such as fiscal year 2026 (filed in 2027), when small companies would be required to begin reporting Scope 3 emissions); Center for Sustainable Business at the University of Pittsburgh (June 17, 2022) ("CSB") (recommending requiring universal disclosure of Scope 3 emissions in 3-5 years of effectiveness of the final rule); and PwC (recommending requiring SRCs that have included Scope 3 emissions in their targets and goals to disclose those emissions).

948 *See, e.g.*, letters from Amer. for Fin. Reform, Sunrise Project *et al*.; and Center Amer. Progress.

949 *See* letters from CalSTRS; Center Amer. Progress; and J. McClellan.

950 *See, e.g.*, letters from Amer. for Fin. Reform, Sunrise Project *et al*.; ICI; and Soros Fund.

951 *See, e.g.*, letters from BIO; Davis Polk; Grant Thornton; D. Burton, Heritage Fdn.; J. Herron; Nasdaq, Inc. (June 14, 2022) ("Nasdaq"); Shearman Sterling; and SBCFAC Recommendation.

rules would likely deter many potential EGCs from going public.[952] Other commenters opposed exempting EGCs from the proposed rules because such companies, like SRCs, may be exposed to climate-related risks.[953]

b. Presentation of the GHG Emissions Metrics and Underlying Methodologies and Assumptions

Commenters expressed mixed views on the proposed requirement to disclose GHG emissions on both an aggregated and disaggregated basis. Some commenters supported the proposed requirement because each constituent gas may be subject to differing regulations and presents its own set of risks, which aggregated disclosure, by itself, would conceal.[954] Other commenters supported the proposed requirement because it would standardize the GHG emissions disclosure and help investors compare the GHG emissions data when making their risk assessments regarding a registrant.[955] Still other commenters supported the proposed requirement because it is consistent with the GHG Protocol and would generally enhance the transparency of GHG emissions disclosure, which they viewed as fundamental for investors because it helps investors understand the financial impacts that transition risk may have on a registrant's business and financial condition, including on its liquidity and capital resources.[956]

Other commenters, however, opposed the proposed requirement to disclose GHG emissions on a disaggregated basis because they believe it would impose additional costs without

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952 *See, e.g.*, letters from Davis Polk; and Grant Thornton.

953 *See, e.g.*, letters from ICI; PwC; and Soros.

954 *See, e.g.*, letters from PwC; and WRI.

955 *See, e.g.*, letters from Amer. for Fin. Reform, Sunrise Project *et al.*; As You Sow; and Wellington Mgmt.

956 *See, e.g.*, letters from Calvert; Fidelity; C. Howard; Impax Asset Mgmt.; and Morningstar.

necessarily resulting in material disclosure.[957] Several of these commenters stated that a registrant should only be required to disclose disaggregated data for constituent gases that are material.[958] Other commenters opposed the proposed requirement because it would be difficult to obtain the necessary data for each constituent gas, particularly for Scopes 2 and 3 emissions.[959] One commenter stated that the proposed disaggregated disclosure requirement would not be compatible with certain industry standard life cycle assessment models.[960] Another commenter opposed a disaggregated disclosure requirement for GHG emissions unless a registrant's particular industry required such disclosure.[961]

Many commenters supported the proposed requirement to describe the methodology, significant inputs, and significant assumptions used to calculate a registrant's GHG emissions metrics.[962] Commenters stated that such disclosure is necessary to place the GHG emissions data in context and to help investors properly understand and interpret the reported emissions information and associated risks.[963] One commenter, however, opposed the proposed requirement, asserting that it would require extensive disclosure of information that is unlikely to be material to investors and will require significant additional effort by registrants.[964] Other commenters opposed a requirement to disclose the emission factors used when calculating GHG

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957 *See, e.g.*, letters from ABA; ERM CVS; Sullivan Cromwell; and T Rowe Price.

958 *See, e.g.*, letters from ABA; Sullivan Cromwell; and T Rowe Price.

959 *See, e.g.*, letters from Cleary Gottlieb; Deloitte & Touche; and Walmart.

960 *See* letter from Amazon.

961 *See* letter from CEMEX.

962 *See, e.g.*, letters from CalPERS; Calvert; Impax Asset Mgmt.; and WRI.

963 *See, e.g.*, letters from CalPERS; and WRI.

964 *See* letter from ABA.

emissions because, in their view, such disclosure would be burdensome to produce and of limited use by investors.[965]

Many commenters stated that a registrant should be required to calculate its GHG emissions pursuant to the GHG Protocol because the GHG Protocol's methodologies have been widely accepted and requiring their adherence would promote comparability.[966] Several of these commenters further recommended that the Commission allow registrants to follow the GHG Protocol's methodology regarding setting organizational boundaries[967] instead of the proposed requirement to base a registrant's organizational boundaries on the entities included in its consolidated financial statements. One of these commenters stated that because many registrants use the "operational control" approach permitted under the GHG Protocol, allowing such registrants to continue to follow the GHG Protocol in this regard would mitigate the compliance burden of GHG emissions disclosure because those registrants would not be required to implement a different approach, in particular, regarding equity method investees.[968] Some commenters, however, stated that a registrant should be permitted to follow other climate-related

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965 *See, e.g.*, letters from ABA; D. Hileman Consulting; ERM CVS; and Futurepast (June 16, 2022).

966 *See, e.g.*, letters from Alphabet *et al.*; As You Sow; Beller *et al.*; CalSTRS; CFA; Dell; Deloitte & Touche; Engine No. 1; ERM CVS; KPMG; Morningstar; Soc. Corp. Gov.; and WRI.

967 *See, e.g.*, letters from Alphabet *et al.*; Beller *et al.*; Deloitte & Touche; and KPMG; *see also* Soc. Corp. Gov (stating that because many registrants use the operational control method, the proposed GHG emissions requirement would not only require unnecessary additional time, effort, and resources and present significant challenges, but it would also generate discrepancies between earlier-reported data and data disclosed pursuant to the proposed rule). *See also* discussion *supra* note 938.

968 *See* letter from Alphabet *et al*.

standards, such as certain International Organization for Standardization (ISO) standards, used by some companies when calculating their GHG emissions.[969]

Several commenters supported the proposed requirement to disclose gross emissions by excluding any purchased or generated carbon offsets.[970] Commenters stated that requiring the disclosure of gross emissions would enable investors to gain a full picture of a registrant's emissions profile and better assess its transition risk exposure.[971] Some commenters also pointed to perceived problems in carbon offset markets regarding the quality and permanence of offsets when supporting a gross emissions disclosure requirement.[972] Other commenters stated that a registrant should be required to disclose both a total amount with, and a total amount without, the use of offsets for each scope of emissions because such disclosure would increase transparency on offset use, which is critical to understanding how a registrant is managing transition risk to its business.[973]

Some commenters, however, opposed the proposed requirement to exclude carbon offsets when disclosing GHG emissions.[974] These commenters stated that the purchase of carbon offsets is a legitimate means for a registrant to reduce its carbon emissions and expressed the view that

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969 *See* letters from Futurepast (referencing ISO 14064-1, Specification with guidance at the organization level for quantification and reporting of greenhouse gas statements and ISO 14067, Carbon footprint of products—Requirements and guidelines for quantification); and International Organization for Standardization (ISO) Committee on GHG and Climate Change Management (June 13, 2022) ("ISO Comm. GHG").

970 *See, e.g.*, letters from AllianceBernstein; Amer. for Fin. Reform, Sunrise Project *et al.*; As You Sow; CalPERS; Etsy; C. Howard; ICCR; KPMG; and Wellington Mgmt.

971 *See, e.g.*, letters from Anthesis Group; As You Sow; CEMEX; Domini Impact; ICI; IATP; KPMG; PRI; and Wellington Mgmt.

972 *See, e.g.*, letters from Amer. For Fin. Reform, Sunrise Project *et al.*; Ceres; and ICCR.

973 *See, e.g.*, letters from AllianceBernstein; CalPERS; and ERM CVS.

974 *See, e.g.*, letters from Airlines for America; International Air Transport Association (June 17, 2022) ("IATA"); and SIFMA (each opposed to a requirement to solely disclose GHG emissions in gross terms and supporting GHG emissions disclosure both in gross and net terms); *see also* letter from J. Weinstein (opposed to any requirement to exclude carbon offsets when disclosing GHG emissions).

high-quality carbon offsets should play a significant role in a transition to a lower carbon economy.[975]

A number of commenters supported the proposed requirement to disclose GHG emissions in terms of GHG intensity.[976] These commenters stated that investors would find the disclosure of GHG intensity useful because it would help them assess a registrant's progress in achieving its emissions management and reduction goals, put in context its emissions in relation to its scale, and facilitate comparing the registrant's emissions efficiency with other registrants in the same industry.[977] Some commenters also noted that the TCFD recommends the disclosure of GHG emissions both in absolute terms and terms of intensity because each metric serves a different purpose.[978] For example, one commenter stated that the disclosure of emissions in absolute terms provides necessary baseline emissions data whereas normalizing the data using an intensity metric allows for a focus on emissions efficiency per unit of production relevant to the registrant's industry.[979] While some commenters supported the proposed requirement to disclose GHG intensity in terms of both metric tons of CO_2e per unit of total revenue and per unit of production relevant to the registrant's industry,[980] other commenters recommended making the

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975 *See* letters from Airlines for America; and SIFMA.

976 *See, e.g.*, letters from Amazon; BOA; CalPERS; D. Hileman Consulting; C. Howard; Morningstar; PIMCO; and PRI.

977 *See, e.g.*, letters from Amazon; BOA; and PIMCO.

978 *See, e.g.*, letters from BOA; and PRI.

979 *See* letter from BOA.

980 *See, e.g.*, letters from Amazon (stating that an intensity metric based on "gross merchandise sales" should be an appropriate unit of production); ERM CVS (stating that an intensity metric based on unit of production should be required where possible); and C. Howard.

final rules more flexible by expressly permitting registrants to use other GHG intensity metrics.[981]

Some commenters, however, opposed the proposed GHG intensity disclosure requirement.[982] These commenters stated that the proposed requirement to disclose a registrant's GHG emissions per unit of total revenue was unnecessary because investors can easily calculate this metric from a registrant's gross GHG emissions divided by its total revenues.[983] Some commenters further stated that the proposed requirement to disclose a registrant's GHG emissions per unit of production would be unworkable for many registrants with different product lines, even within the same industry, and would not result in comparable disclosure for investors.[984] Consequently, according to these commenters, GHG intensity disclosure should only be voluntary.[985]

Several commenters supported the proposed provision that would allow a registrant to use reasonable estimates when disclosing its GHG emissions as long as it also describes the assumptions underlying, and its reasons for using, the estimates.[986] One commenter stated that the proposed provision would encourage the disclosure of GHG emissions.[987] Other commenters supported the proposed provision because the reporting of GHG emissions often relies on the use

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981 *See, e.g.*, letters from BOA (stating that registrants should be permitted to use GHG intensity metrics specified under the TCFD framework or incorporated into the Partnership for Carbon Accounting Financials' Global GHG Accounting & Reporting Standard used by banks and other financial institutions); and NAM (supporting increased flexibility that would allow companies to choose and disclose a single GHG intensity metric, or to forgo intensity reporting, depending on the metrics' relevance to their operations and emissions).

982 *See, e.g.*, letters from ABA; PwC; SIFMA; and Sullivan Cromwell.

983 *See* letters from ABA; and Sullivan Cromwell.

984 *See* letters from ABA; PwC; SIFMA; and Sullivan Cromwell.

985 *See, e.g.*, letters from CEMEX; PwC; and SIFMA.

986 *See, e.g.*, letters from C2ES; CEMEX; D. Hileman Consulting; ERM CVS; KPMG; PWC; and WSP.

987 *See* letter from Cemex.